CS/EE 147 – GPU Computing and Programming

Introduction

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Welcome to CS/EE 147!

Ice Breaker – About me

I moved here from Bangladesh in 2021

Find it on the map

Education

Bangladesh University of Engineering and Technology

BS Computer Science and Engineering ’19

University of California Riverside

MS Computer Science ’25

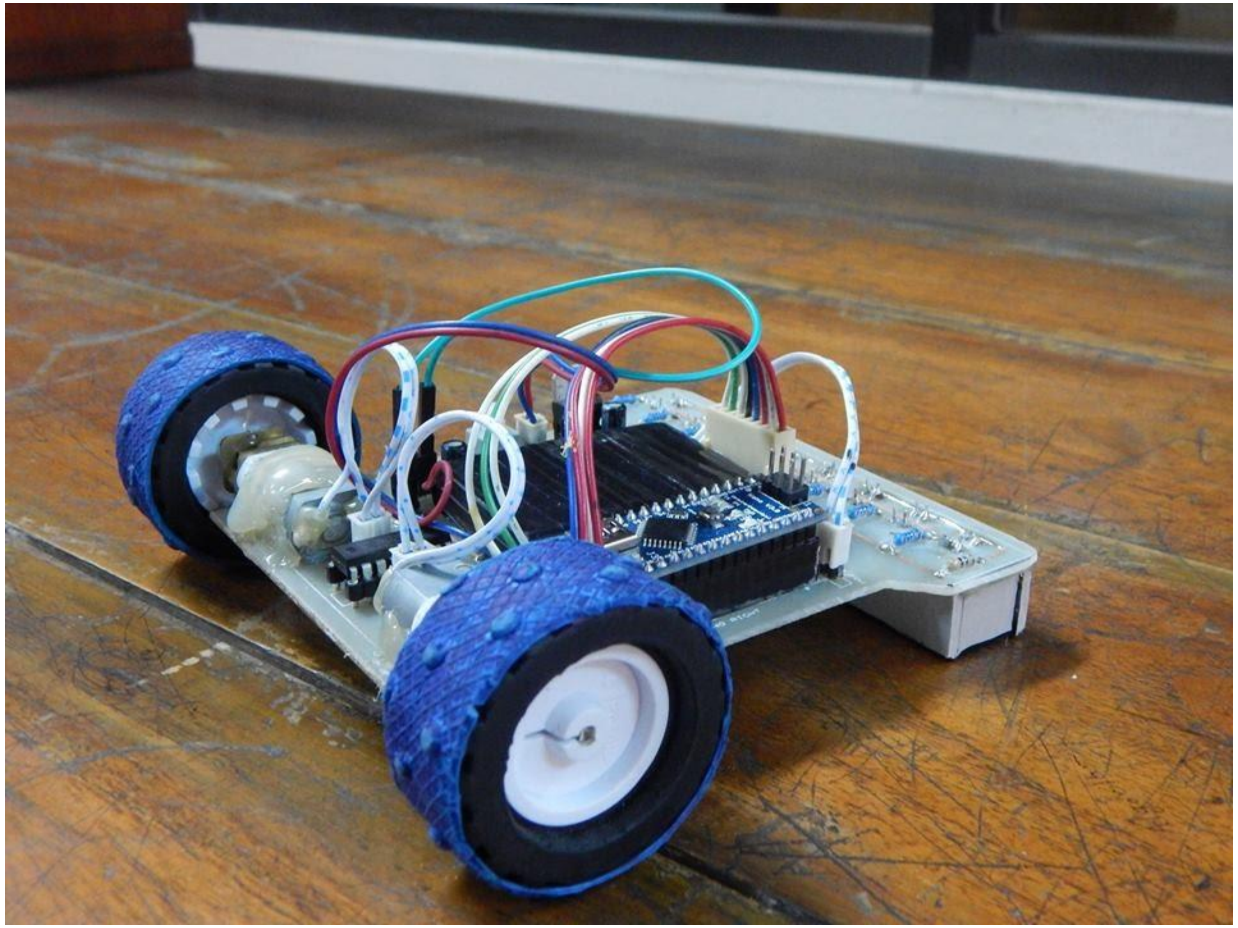
University of California Riverside

PhD Candidate

SOCAL research group (led by Prof Daniel Wong)

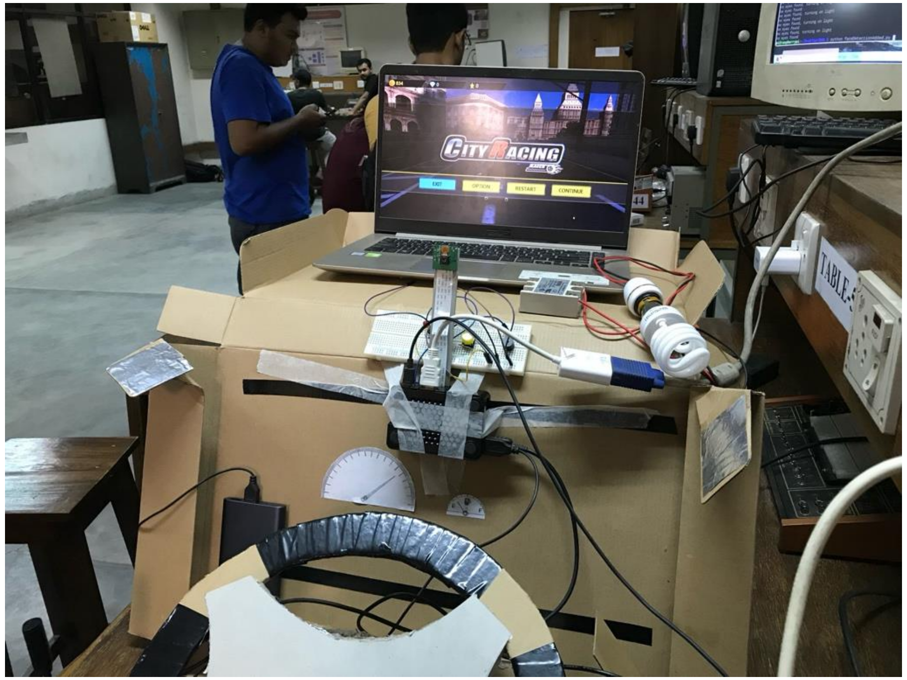
Why Computer Architecture?

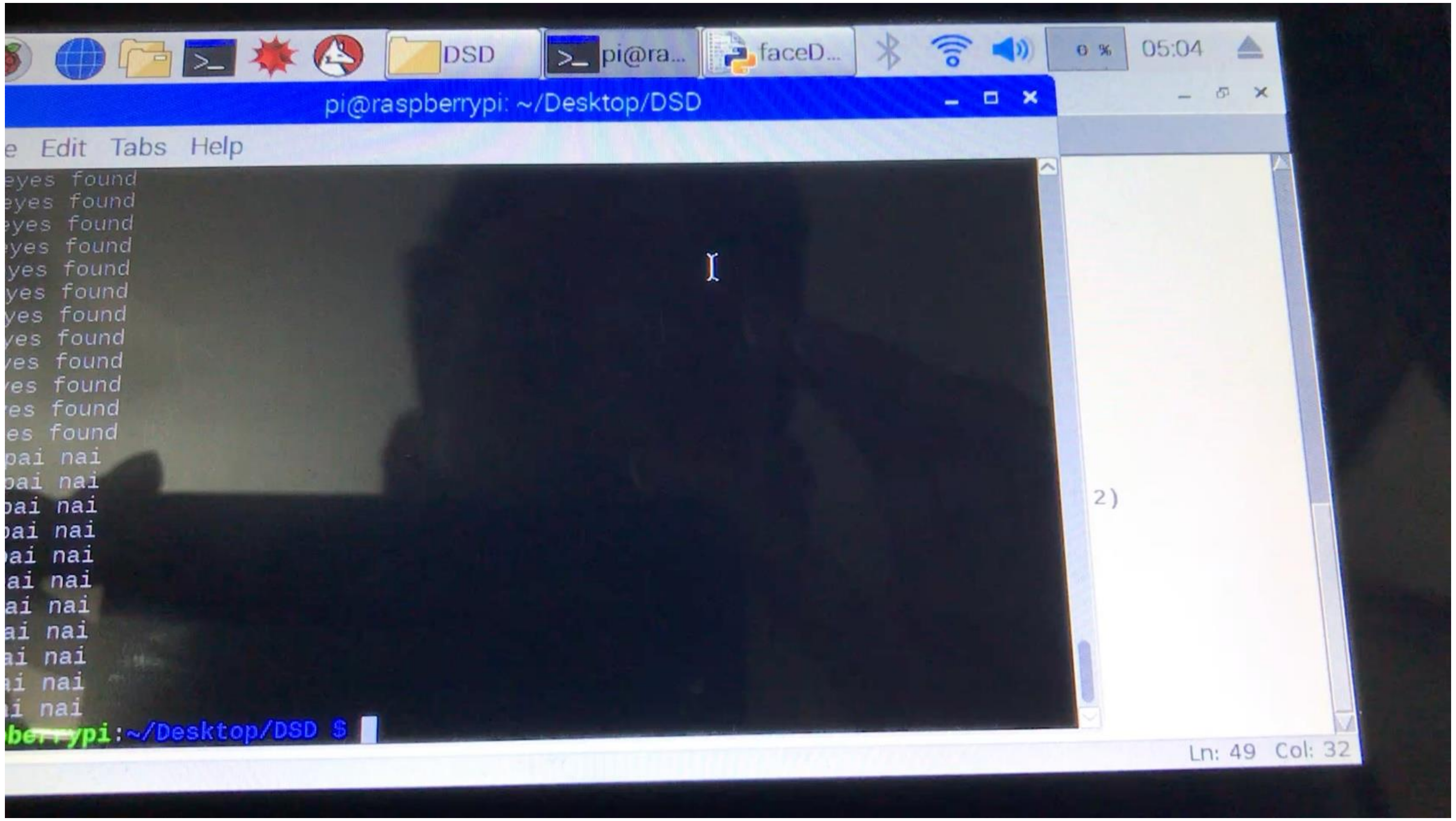
Robotics



Why Computer Architecture?

Image processing





Why Computer Architecture?

Undergrad thesis in enabling GPGPU Frameworks for HSA capable APUs (don’t ask, this is very outdated now)

Why Computer Architecture?

CSRankings: Computer Science Rankings

CSRankings is a metrics-based ranking of top computer science institutions based on faculty publications at selective conferences.

USA | 2016 | 2026 | All | AI | Systems | Theory | Interdisc. | Tour | Sponsor CSrankings

All Areas

AI
- ▶ Artificial intelligence ☐
- ▶ Computer vision ☐
- ▶ Machine learning ☐
- ▶ Natural language processing ☐
- ▶ The Web & information retrieval ☐

Systems
- ▶ Computer architecture ☑
- ▶ Computer networks ☐
- ▶ Computer security ☐

	Institution	Count	Faculty
1	▶ Univ. of Illinois at Urbana-Champaign	42.3	23
2	▶ University of Michigan	34.2	25
3	▶ Georgia Institute of Technology	29.2	17
4	▶ Massachusetts Inst. of Technology	26.8	15
5	▶ Univ. of California - San Diego	21.2	20
6	▶ Pennsylvania State University	18.2	16
7	▶ Univ. of California - Riverside	17.6	12
8	▶ Carnegie Mellon University	17.4	22
9	▶ North Carolina State University	17.1	13

CS/EE 147
GPU COMPUTING AND PROGRAMMING

Contact

- Instructor: Nafis Mustakin
  - Email: nmust004@ucr.edu
  - Homepage: nmustakin.github.io
  - Office Hours: Wednesday 1-2PM @ WCH110

- TA: Sara Rashidi Golrouye
  - Email: srash034@ucr.edu
  - Office Hours: TBD

Course Goals

- Learn how to program GPGPU processors and achieve
  - High performance
  - Functionality and maintainability
- Learn how GPGPUs work
  - Microarchitecture
  - Design challenges
- Technical subjects
  - principles and patterns of parallel algorithms
  - processor architecture features and constraints
  - programming API, tools and techniques

Logistics

- Course Website
  - http://nmustakin.github.io/teaching/CSEE147Sp26
  - Check often for announcements
- eLearn (eLearn.ucr.edu)
  - Grades
  - Yuja for hosting recorded lecture videos
- Slack for discussion (Find link on eLearn)
- ENGR Account Setup
  https://www.engr.ucr.edu/secured/systems/login.php

Textbook (Not really required…)

1. D. Kirk and W. Hwu, "Programming Massively Parallel Processors – A Hands-on Approach, Second Edition"

2. CUDA by example, *Sanders and Kandrot*

3. *Nvidia CUDA C Programming Guide*
   - *https://docs.nvidia.com/cuda/cuda-c-programming-guide/*

Attendance/Recordings

- Attendance
  - You are expected to attend all lectures.
  - However, …stay home if you're sick or feeling unwell. ☺

- <u>You</u> are responsible for keeping up with course lectures and assignments!

Grading

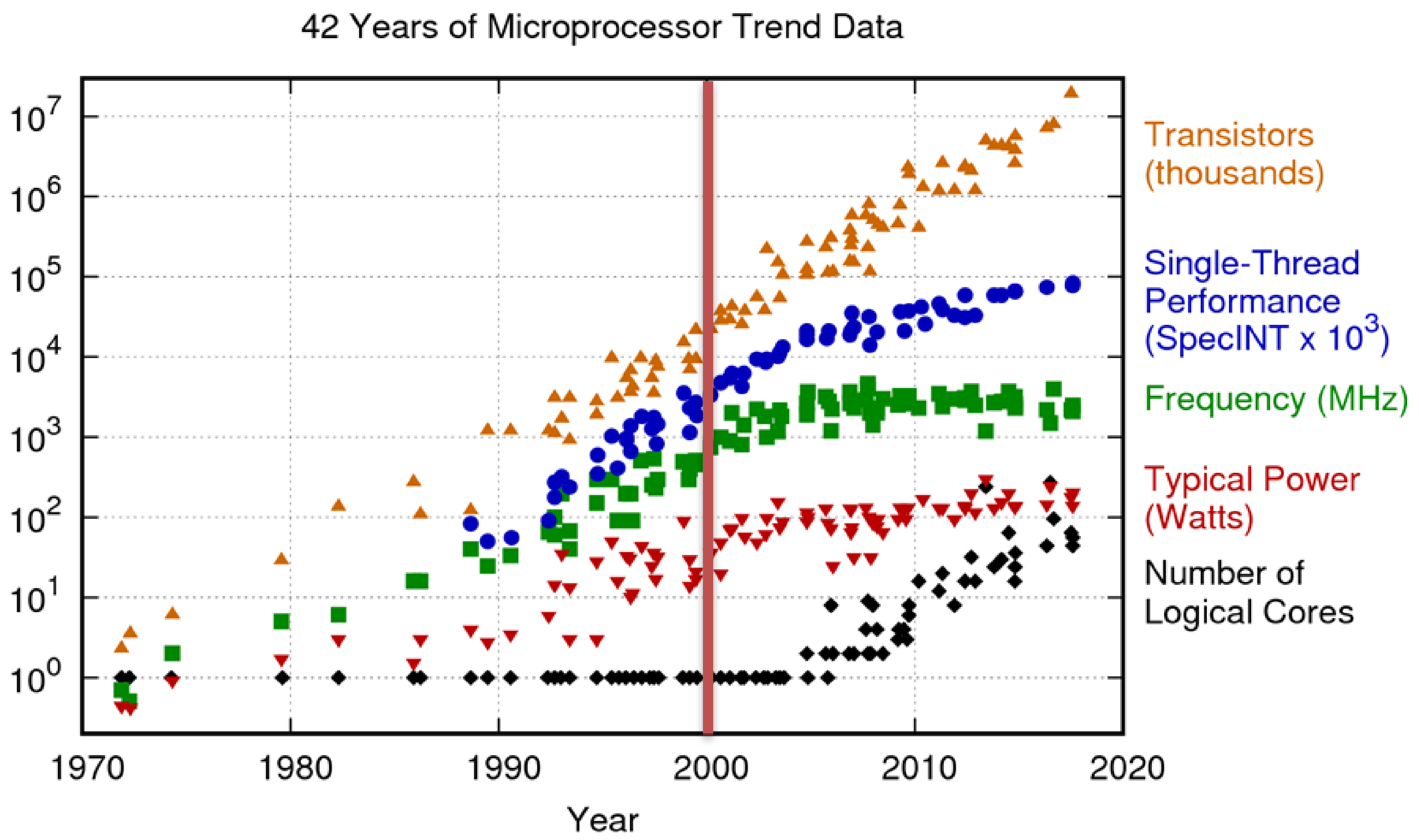
- Grade Breakdown
  - Assignments: 28% (4 assignments, 7% each)
  - Final Project: 25%
  - Exam 1: 25%
  - Exam 2: 20%
  - Class Participation / Discussion: 2%

Assignment Policies

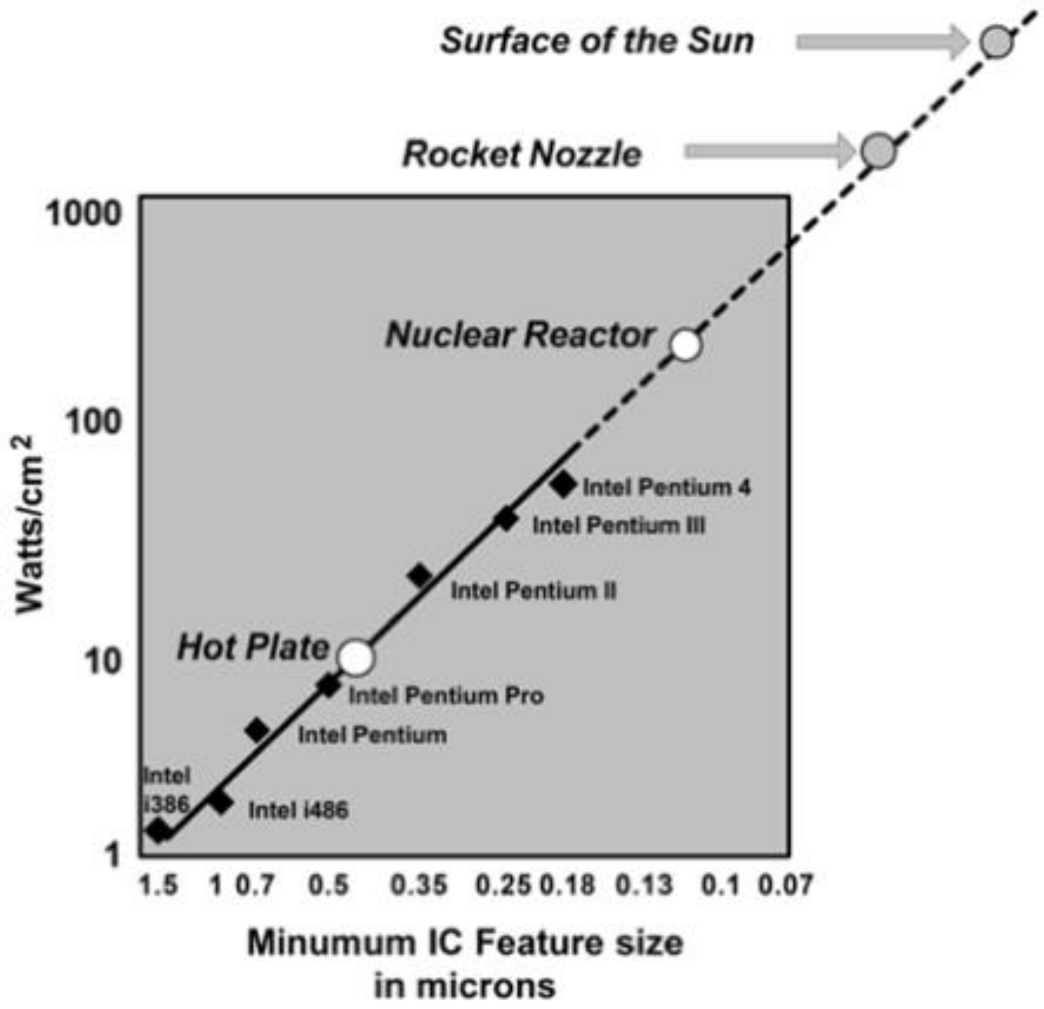
- 3 slip days
- 15% penalty per late day
- If it's one minute late, it's still late
- No extensions will be given
- All assignments/projects are due at the end of the due date (midnight Pacific time)

QUESTIONS?

Why GPUs?



Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp


Surface of the Sun
Rocket Nozzle
1000
100
10
1
Watts/cm 2
Nuclear Reactor
Intel Pentium 4
Intel Pentium III
Intel Pentium II
Hot Plate
Intel Pentium Pro
Intel Pentium
Intel i386
Intel i486
1.5 1 0.7 0.5 0.35 0.25 0.18 0.13 0.1 0.07
Minumum IC Feature size in microns

Modern computer architecture is limited by:

a. Process Technology / Transistor density
   (End of Moore's Law)

b. Power
   (End of Dennard Scaling)

c. Temperature

General-purpose
(Easier to program)

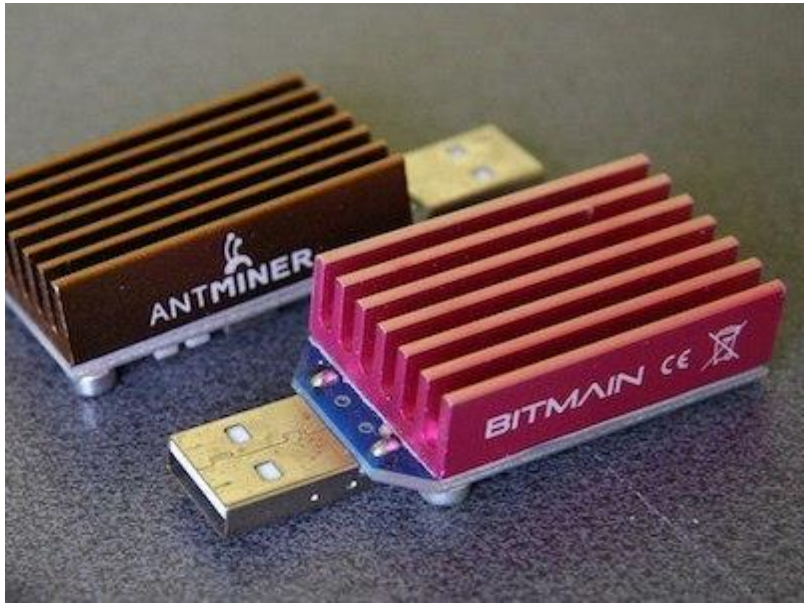
Single-core CPU | Multi-core CPU | GPU | FPGA | ASIC

Specialized

Energy-efficient

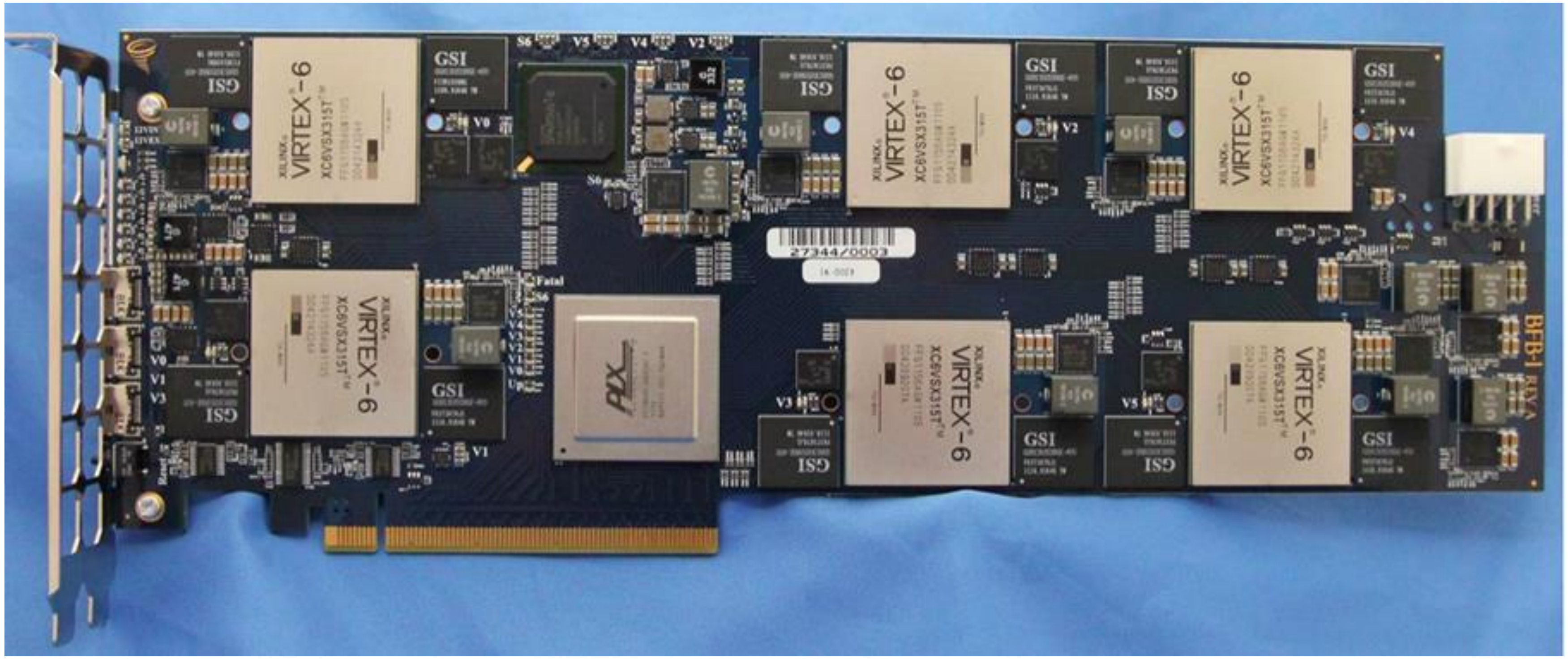
Examples of Specialization

Bitcoin Mining



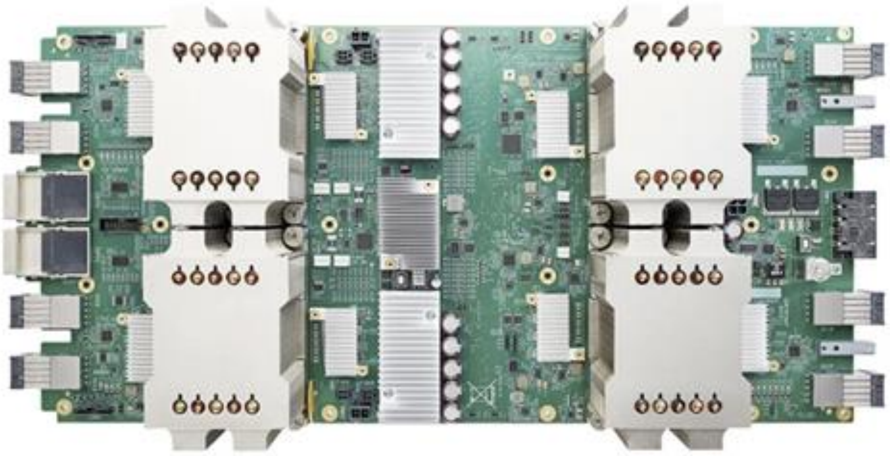
Examples of Specialization

Microsoft Catapult

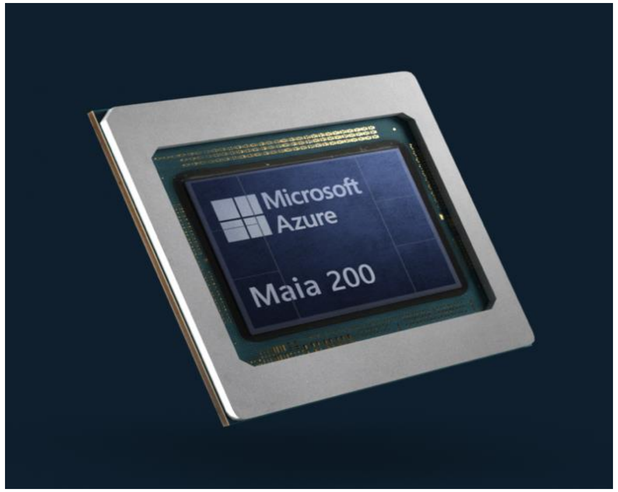


Examples of Specialization

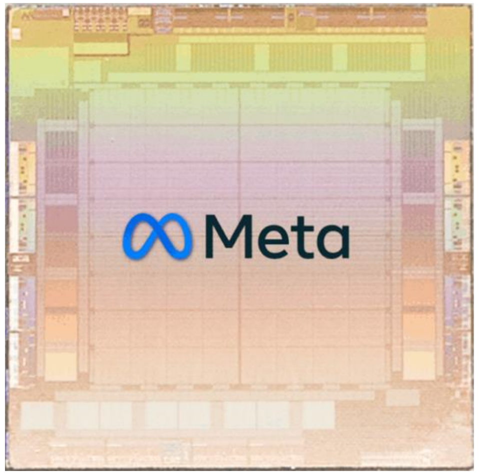
Google TPU - Tensor Processing Unit



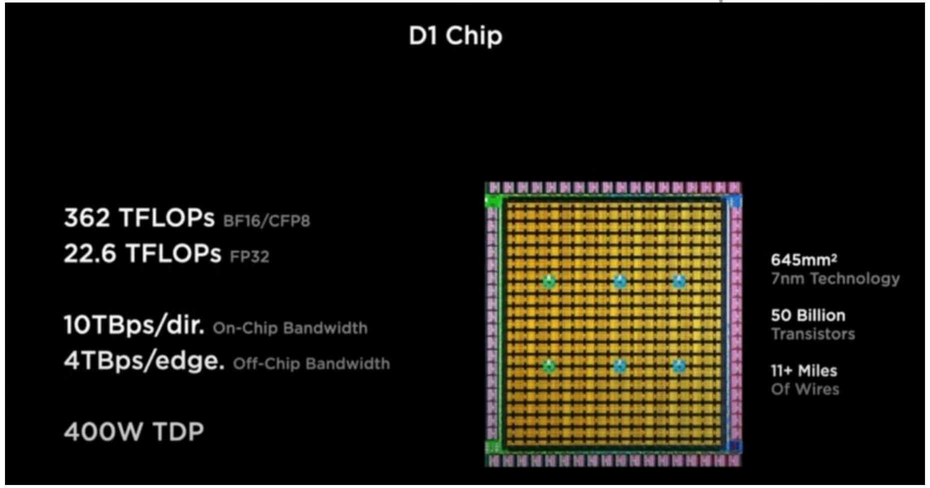
Examples of Specialization



Microsoft Maia



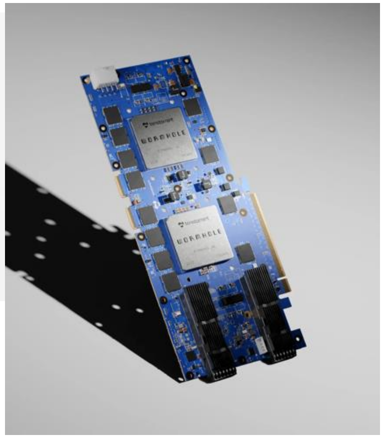
Meta MTIA



Tesla D1

Baidu Kunlun II

Graphcore



Tenstorrent

SambaNova

Cerebras

Examples of Specialization

GPUs - Supercomputers

Rank	System	Cores	Rmax (PFlop/s)	Rpeak (PFlop/s)	Power (kW)
1	**El Capitan** - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, TOSS, HPE DOE/NNSA/LLNL United States	11,039,616	1,742.00	2,746.38	29,581
2	**Frontier** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, HPE Cray OS, HPE DOE/SC/Oak Ridge National Laboratory United States	9,066,176	1,353.00	2,055.72	24,607
3	**Aurora** - HPE Cray EX - Intel Exascale Compute Blade, Xeon CPU Max 9470 52C 2.4GHz, Intel Data Center GPU Max, Slingshot-11, Intel DOE/SC/Argonne National Laboratory United States	9,264,128	1,012.00	1,980.01	38,698
4	**Eagle** - Microsoft NDv5, Xeon Platinum 8480C 48C 2GHz, NVIDIA H100, NVIDIA Infiniband NDR, Microsoft Azure Microsoft Azure United States	2,073,600	561.20	846.84	
5	**HPC6** - HPE Cray EX235a, AMD Optimized 3rd Generation EPYC 64C 2GHz, AMD Instinct MI250X, Slingshot-11, RHEL 8.9, HPE Eni S.p.A. Italy	3,143,520	477.90	606.97	8,461
6	**Supercomputer Fugaku** - Supercomputer Fugaku, A64FX 48C 2.2GHz, Tofu interconnect D, Fujitsu RIKEN Center for Computational Science Japan	7,630,848	442.01	537.21	29,899
7	**Alps** - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE Cray OS, HPE Swiss National Supercomputing Centre (CSCS) Switzerland	2,121,600	434.90	574.84	7,124

MACHINE LEARNING! AI!

ML/AI is pervasive

UCR

Gemini

deepseek

cohere

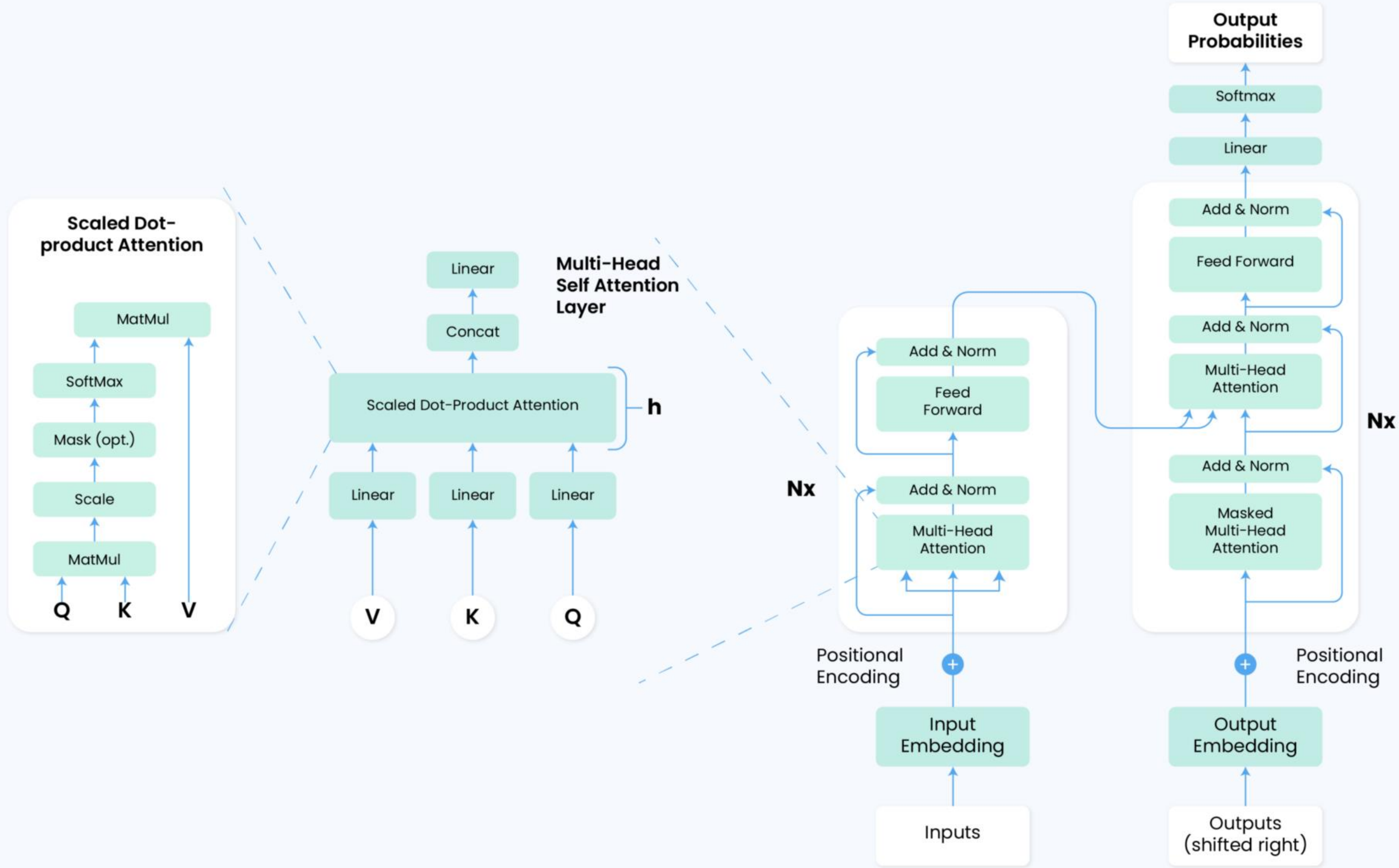
OpenAI

perplexity

stability.ai

ANTHROPIC

databricks



Scaled Dot-product Attention
MatMul
SoftMax
Mask (opt.)
Scale
MatMul
Q
K
V
Linear
Concat
Multi-Head Self Attention Layer
Scaled Dot-Product Attention
h
Linear
Linear
Linear
V
K
Q
Nx
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Positional Encoding
Input Embedding
Inputs
Output Probabilities
Softmax
Linear
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Nx
Add & Norm
Masked Multi-Head Attention
Positional Encoding
Output Embedding
Outputs (shifted right)

MM Support on GPU

cuBLAS

NVIDIA/cutlass

CUDA Templates for Linear Algebra Subroutines

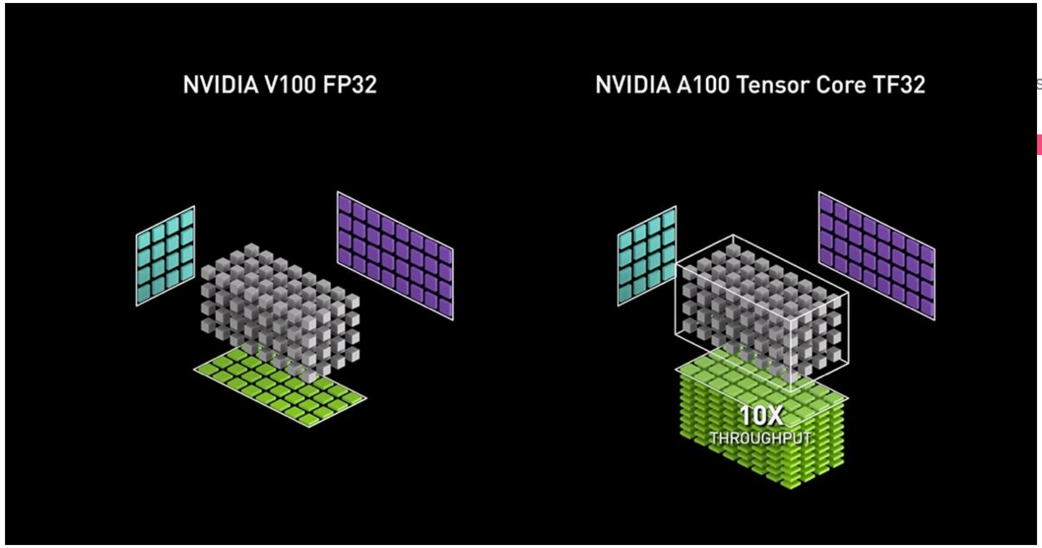
7k Stars

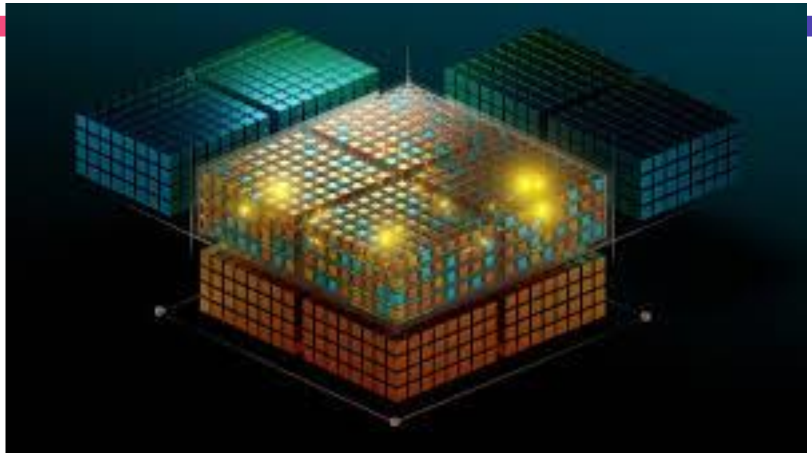
ROCm/rocBLAS

Next generation BLAS implementation for ROCm platform

AMD ROCm

109 Contributors | 1 Issue | 362 Stars | 175 Forks





AMD Matrix Cores

ML+GPU Ecosystem

UCR

ML Frameworks

- PyTorch
- TensorFlow
- JAX

Language & Compilers

- Modular
- OpenAI Triton
- OctoAI®

Runtime

- NVIDIA® CUDA®
- AMD ROCm
- OpenCL™

QUESTIONS?